

Lesson 14 Classification of Coxeter graph

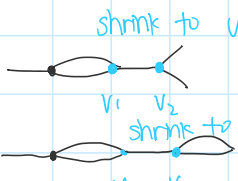
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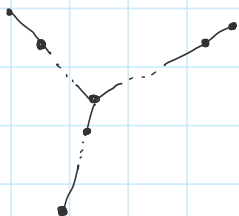
Classification of Coxeter graphs

From last time, for any v_i , # of edges adjacent to v_i is ≤ 4 .

Thus we examine possible subgraphs:

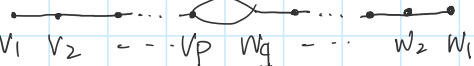
1°  $\Rightarrow$ the whole graph.

2°  $\Rightarrow \dots \Rightarrow$  $\times \Rightarrow \dots$

3°  or 

Thus we have

1° 

2° 

v and w are l.i.

We consider $v = v_1 + 2v_2 + \dots + pv_p$
 $w = w_1 + 2w_2 + \dots + qw_q$

Note that $(v|w)^2 \leq (v|v)(w|w)$

Calculate:

$$\begin{aligned} (v|v) &= 1^2 + 2^2 + \dots + p^2 - 2 \cdot \frac{1}{2} \cdot (1 \cdot 2 + 2 \cdot 3 + \dots + (p-1)p) \\ &= p^2 - \frac{(p-1)p}{2} \\ &= \frac{p(p+1)}{2} \end{aligned}$$

$$(w|w) = \frac{q(q+1)}{2}$$

$$(v|w) = pq - \frac{\sqrt{2}}{2}$$

$$\begin{aligned} \text{Thus we have } \frac{1}{2} p^2 q^2 &< \frac{pq(p+1)(q+1)}{4} \\ 2pq &< (p+1)(q+1) \end{aligned}$$

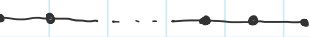
$$pq - p - q + 1 < 2$$

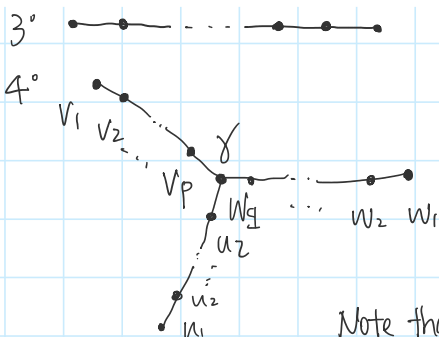
$$(p-1)(q-1) < 2$$

$$\Rightarrow \begin{cases} p=1 \text{ or } q=1 \\ p \geq 2, q \geq 2 \Rightarrow p=q=2 \end{cases}$$

i.e. 

 ($q=1$ is equivalent to $p=1$ by permutation)

3° 



Consider

$$V = v_1 + 2v_2 + \dots + pv_p$$

$$W = w_1 + 2w_2 + \dots + qw_q$$

$$U = u_1 + 2u_2 + \dots + ru_r$$

Note that $(\gamma | \frac{V}{\|V\|})^2 + (\gamma | \frac{W}{\|W\|})^2 + (\gamma | \frac{U}{\|U\|})^2 < 1$

i.e. $\frac{(\gamma | V)^2}{(V | V)} + \frac{(\gamma | W)^2}{(W | W)} + \frac{(\gamma | U)^2}{(U | U)} < 1$

(note that V, W, U are pairwise orthogonal, we may extend $\frac{V}{\|V\|}, \frac{W}{\|W\|}, \frac{U}{\|U\|}$ to an orthonormal basis, then $\gamma = (\gamma | \frac{V}{\|V\|}), \dots$ and the inequality follows from $\|\gamma\|=1$)

We have $(\gamma | V) = -\frac{p}{2}$
 $(V | V) = \frac{p(p+1)}{2} \Rightarrow \frac{(\gamma | V)^2}{(V | V)} = \frac{\frac{p^2}{4}}{\frac{p(p+1)}{2}} = \frac{1}{2} \frac{p}{p+1}$

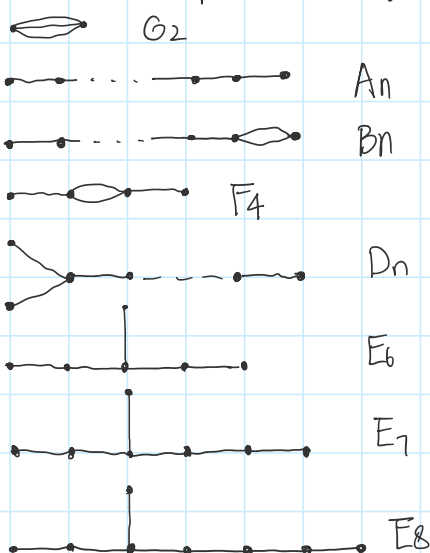
Thus $\frac{1}{2} \left(1 - \frac{1}{p+1} + 1 - \frac{1}{q+1} + 1 - \frac{1}{r+1} \right) < 1$

$$\frac{1}{p+1} + \frac{1}{q+1} + \frac{1}{r+1} > 1$$

Assume $p \leq q \leq r \Rightarrow p=1 \Rightarrow \frac{1}{q+1} + \frac{1}{r+1} > \frac{1}{2}$

$\begin{cases} q=1 & r \text{ can be arbitrary} \\ q=2 & \frac{1}{r+1} > \frac{1}{6} \Rightarrow r+1 < 6 \Rightarrow r=2, 3, 4 \end{cases}$

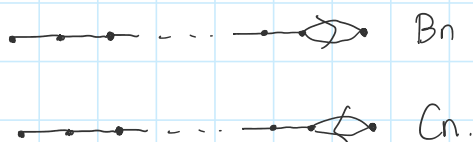
We have the complete classification:



Def Dynkin diagram

Whenever a double or triple edge occurs in the Coxeter graph of Φ , we can add an arrow pointing to the shorter of the two roots. We call the resulting figure the Dynkin diagram of Φ .

Then



Realization of root systems

Let $E = \mathbb{R}^n$ with the usual inner product and $w_1, \dots, w_n$ denote the usual orthonormal unit vectors which form a basis of $\mathbb{R}^n$.

The $\mathbb{Z}$ -span of this basis is by definition of a lattice denoted Γ . The root system Δ will be defined to be the set of all vectors in Γ having specified length or lengths.

Rmk roughly check that Δ is a root system.

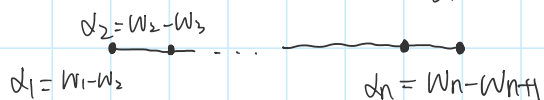
By def, it is easy to see that Δ is finite, spans E and exclude 0; $\forall \alpha \in \Delta, \exists \lambda \in \mathbb{Z} \setminus \{0\} : \lambda \alpha \in \Delta$; to satisfy $\forall \alpha, \beta \in \Delta, \langle \alpha, \beta \rangle \in \mathbb{Z}$ it usually suffices to choose squared lengths dividing 2; for the reflection s_α , it is enough to check that $s_\alpha(\alpha \in \Delta)$ maps Δ back to Γ since then $s_\alpha(\Delta)$ automatically consists of vectors of the required lengths.

A_n

Let $E = \mathbb{R}^n$ be n -dimensional subspace of $\mathbb{R}^{n+1}$ orthogonal to the vector $w_1 + \dots + w_{n+1}$. Let $\Gamma' = \Gamma \cap E = \{ \sum_{i=1}^{n+1} k_i w_i \mid \sum k_i = 0, k_i \in \mathbb{Z} \}$

$$A_n = \{ \alpha \in \Gamma' \mid (\alpha|\alpha) = 2 \} \Rightarrow A_n = \{ w_i - w_j \mid 1 \leq i \neq j \leq n+1 \}$$

The vectors $w_1 - w_2, w_2 - w_3, \dots, w_n - w_{n+1}$ form a basis B of A_n , i.e.



$$\Delta^+ = \{ w_i - w_j \mid i < j \}$$

$$\Delta^- = \{ w_i - w_j \mid i > j \}$$

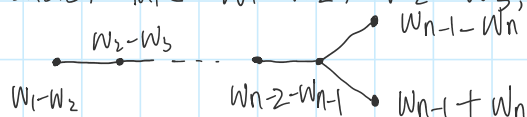
Notice that $s_{\alpha_i} = w_i - w_{i+1}$ permutes the subscripts $i, i+1$ and leaves other subscripts fixed $\Rightarrow s_{\alpha_i}$ corresponds to the transposition $(i \ i+1)$ in S_{n+1} .

From $W = \langle B \rangle$, we conclude that $W = S_{n+1}$.

D_n

Let $E = \mathbb{R}^n$ and $D_n = \{ \alpha \in \Gamma \mid (\alpha|\alpha) = 2 \} \Rightarrow D_n = \{ \pm w_i \pm w_j \mid i \neq j \}$.

For a basis, take $w_1 - w_2, w_2 - w_3, \dots, w_{n-1} - w_n, w_{n-1} + w_n$, i.e.



B_n

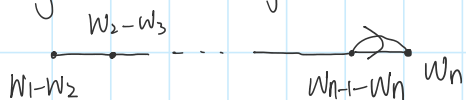
Let $E = \mathbb{R}^n$ and $B_n = \{ \alpha \in \Gamma \mid (\alpha|\alpha) = 1 \text{ or } 2 \} \Rightarrow B_n = \{ \pm w_i, \pm w_i \pm w_j \mid i \neq j \}$

B_n
 Let $E = \mathbb{R}^n$ and $B_n = \{ \alpha \in \mathbb{I} \mid (\alpha|\alpha) = 1 \text{ or } 2 \} \Rightarrow B_n = \{ \pm w_i, \pm w_i \pm w_j \mid i \neq j \}$

For a basis, take $w_1 - w_2, w_2 - w_3, \dots, w_{n-1} - w_n, w_n$

(note that a short root $w_i = (w_i - w_{i+1}) + (w_{i+1} - w_{i+2}) + \dots + (w_{n-1} - w_n) + w_n$

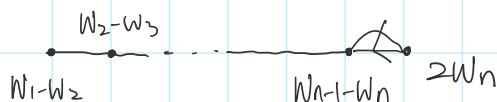
a long root $w_i - w_j$ and $w_i + w_j$ is similarly expressible)



$$\alpha^\vee = \frac{2\alpha}{(\alpha|\alpha)}$$

C_n
 C_n is the root system dual to $B_n \Rightarrow C_n = \{ \pm 2w_i, \pm w_i \pm w_j \mid i \neq j \}$.

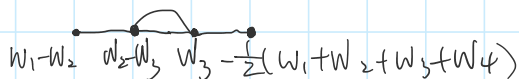
For a basis, take $w_1 - w_2, \dots, w_{n-1} - w_n, 2w_n$



F₄
 Let $E = \mathbb{R}^4$ and $\mathbb{I}' = \mathbb{I} + \frac{\mathbb{Z}}{2}(w_1 + w_2 + w_3 + w_4)$.

$F_4 = \{ \alpha \in \mathbb{I}' \mid (\alpha|\alpha) = 1 \text{ or } 2 \} \Rightarrow F_4 = \{ \pm w_i, \pm w_i \pm w_j \ (i \neq j), \frac{1}{2}(\pm w_1 \pm w_2 \pm w_3 \pm w_4) \}$

For a basis, take $w_1 - w_2, w_2 - w_3, w_3, -\frac{1}{2}(w_1 + w_2 + w_3 + w_4)$



G₂
 $\mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O}$
 1 2 4 8

$G_2 = \text{Der}(\mathbb{O})$ (In fact $\text{Der}(\mathbb{O}_{\mathbb{C}} = \mathbb{O} \otimes_{\mathbb{R}} \mathbb{C})$ gives $G_2(\mathbb{C})$)

Abstractly, we take $E = \mathbb{R}^2$ to be the subspace of $\mathbb{R}^3$ orthogonal to $w_1 + w_2 + w_3$

$G_2 = \{ \alpha \in \mathbb{I} \mid (\alpha|\alpha) = 2 \text{ or } 6 \} \Rightarrow G_2 = \{ w_1 - w_2, w_2 - w_3, w_1 - w_3, 2w_1 - w_2 - w_3, 2w_2 - w_1 - w_3, 2w_3 - w_1 - w_2 \}$

For a base, take $w_1 - w_2, -2w_1 + w_2 + w_3$

